

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 05 / 01 /2026**
Time: 3 Hrs **GTM - 09** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The position of a particle moving along X-axis is given by $x(t) = Ae^{\frac{B\cos t}{t}} + C\log\left(\frac{D}{t}\right)$, dimensions of $\frac{ABD}{C} = M^a L^b T^c$ then $a+b+c=?$

A) 3 B) 2 C) 0 D) 1

2. A pendulum bob of density ρ is oscillating in a liquid of density σ, then frequency of oscillation would be

A) $f = \frac{1}{2\pi}\sqrt{\frac{\left(1-\frac{\sigma}{\rho}\right)g}{l}}$ B) $f = \frac{1}{2\pi}\sqrt{\frac{\left(1+\frac{\sigma}{\rho}\right)g}{l}}$

C) $f = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$

D) $f \rightarrow$ does not depends on relative density

For More Join: @IITJEE_Advanc

3. **Statement - I**: Water rises in a glass capillary tube, but mercury falls in the same tube

Statement - II: The height of the capillary rise is given by the formula $h=\frac{2T\cos\theta}{\rho gr}$, where the factors T (surface tension), θ (angle of contact), ρ (density), g (gravity), and r (capillary radius)

In the light above statements, choose the correct answer from the options given below

A) Statement-I is false but Statement-II is true

B) Statement-I is true but Statement-II is false

C) Both Statement-I and Statement-II are true

D) Both Statement-I and Statement-II are false

4. De-broglie wavelength for proton accelerated to potential V_0 is equal to λ_0. The corresponding De-Broglie wavelength for $\alpha-$particle under similar condition would be

A) $\frac{\lambda_0}{2}$ B) $\frac{\lambda_0}{2\sqrt{2}}$ C) $\frac{\lambda_0}{\sqrt{2}}$ D) $\frac{\lambda_0}{3\sqrt{2}}$

5. When the angle of incidence on a material is 60^0, the reflected light is completely polarized. The velocity of the refracted ray inside the material is $\left(\text{in } ms^{-1}\right)$

A) 3×10^8 B) $\left(\frac{3}{\sqrt{2}}\right)\times10^8$ C) $\sqrt{3}\times10^8$ D) 0.5×10^8

6. When ${}_{92}U^{235}$ undergoes fission. About 0.1% of the original mass is converted into energy. Then the amount of ${}_{92}U^{235}$ should undergo fission per day in a nuclear reactor so that it provides 200 mega watt electric power per day is__

A) $9.6\times10^{-2}kg$ B) $4.8\times10^{-2}kg$

C) $19.2\times10^{-2}kg$ D) $1.2\times10^{-2}kg$

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7. The power factor of an L – R circuit is $\frac{1}{\sqrt{2}}$. If the frequency of AC is doubled, then what will be the power factor?

A) $\frac{1}{\sqrt{3}}$ B) $\frac{1}{\sqrt{5}}$ C) $\frac{1}{\sqrt{7}}$ D) $\frac{1}{\sqrt{11}}$

8. The self-inductance of a coil, filled with medium of Relative permeability (μ_r) is L then, which of the following are correct

a) L becomes $\frac{1}{4}$th when number of turns becomes half

b) L becomes doubled when number of turns are doubled and area of cross-section become half

c) L is independent of medium filled in it

d) Work must be done against induced emf to establish current in it

A) a,b only B) b,c,d only C) a,b,d only D) c,d only

9. Match the thermodynamic processes taking place in a system with the correct conditions. In the table: ΔQ is the heat supplied, ΔW is the work done and ΔU is change in internal energy of the system

Process	Condition
I) Adiabatic	A) $\Delta W = 0$
II) Isothermal	B) $\Delta Q = 0$
III) Isochoric	C) $\Delta U \neq 0, \Delta W \neq 0, \Delta Q \neq 0$
IV) Isobaric	D) $\Delta U = 0$

A) $(I)-(B),(II)-(D),(III)-(A),(IV)-(C)$

B) $(I)-(A),(II)-(A),(III)-(B),(IV)-(C)$

C) $(I)-(A),(II)-(B),(III)-(D),(IV)-(D)$

D) $(I)-(B),(II)-(A),(III)-(D),(IV)-(C)$

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10. Which of the following statements is correct for a moving coil galvanometer?"

i) The deflection is directly proportional to the square of current

ii) The deflection is directly proportional to the area of the coil

iii) The deflection is inversely proportional to the torsional constant of the suspension fibre.

A) i, ii B) ii, iii C) i, iii D) i, ii, iii

11. A short electric dipole is placed at the origin 'O' and is directed along the +ve x-axis. At a point 'P' far away from the dipole, the electric field is parallel to the y-axis. The line 'OP' makes an angle 'θ' with the $x-axis$. Then

A) $\tan\theta = \sqrt{3}$ B) $\tan\theta = \sqrt{2}$ C) $\theta = 45^0$ D) $\tan\theta = \frac{1}{\sqrt{2}}$

12. A metal ball of mass 0.1 kg is heated upto $500^0 C$ and dropped into a vessel of heat capacity 800 J K^{-1} and containing 0.5 kg water. The initial temperature of water and vessel is 30^0C. What is the approximate percentage increment in the temperature of the water?

[specific heat capacities of water and metal are, respectively 4200 J $kg^{-1}K^{-1}$ and 400 J $kg^{-1}K^{-1}$]

A) 15% (approx) B) 30% (approx) C) 25% (approx) D) 21% (approx.)

13. What is lateral shift in the path of ray, if it is incident on a slab of $\mu = \sqrt{3}$ with an angle of incidence 60^0 from the air. Thickness of slab is $= 10cm$

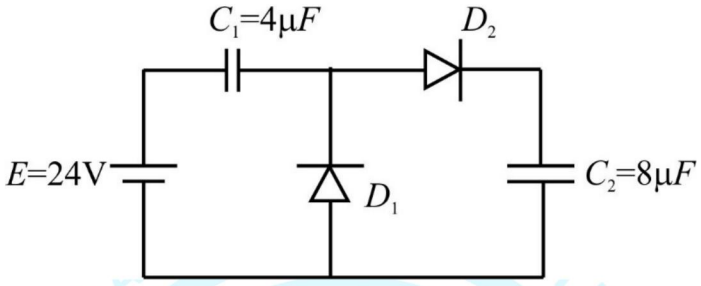
A) $\frac{20}{\sqrt{3}} cm$ B) $\frac{5}{\sqrt{2}} cm$ C) $\frac{10}{\sqrt{3}} cm$ D) $\frac{10}{\sqrt{2}} cm$

14. A ring of mass 'm' and radius 'r' is allowed to roll without slipping from the highest point of an inclined plane of length 'L' and makes an angle 30^0 with the horizontal. The speed of the ring at the bottom of the plane is v_1. If the angle of inclination is increased to 60^0 while keeping L constant. Then the new speed of the ring at the bottom of the plane is v_2. The ratio of $v_1^2 : v_2^2$ is

A) $1:\sqrt{2}$ B) $\frac{\sqrt{3}}{2}$ C) $\frac{2}{\sqrt{3}}$ D) $1:\sqrt{3}$

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15. In the given circuit potential across capacitor C_1, C_2 are V_1, V_2 and diodes being ideal which of the following options may be correct



i) $V_1 = 16V, Q_1 = 64\mu C$ ii) $V_2 = 8V, Q_2 = 64\mu C$

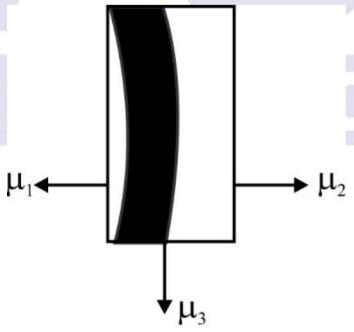
iii) $V_1 = 8V, Q_1 = 64\mu C$ iv) $V_2 = 16V, Q_2 = 96\mu C$

A) i, ii B) ii, iii C) i, iii D) i, iv

16. A force is represented by $F = ax^2 + bt^{\frac{1}{2}}$ where x = distance and t = time. The dimensions of $\frac{b^2}{a}$

A) $\left[ML^2T^{-3}\right]$ B) $\left[ML^3T^{-3}\right]$ C) $\left[MLT^{-2}\right]$ D) $\left[ML^{-1}T^{-1}\right]$

17. The figure given below shows a thin plano-convex lens of refractive index μ_1 and a thin plano-concave lens of refractive index μ_2 both having same radius of curvature R of their curved surfaces. Another thin lens of refractive index μ_3 has same radius of curvature R on the two surfaces between the plano convex and plano concave lenses that the plane surfaces are parallel to each other. Find the focal length of the combination



A) $f = \frac{R}{\mu_1 \mu_2}$ B) $\frac{R}{\mu_1 - \mu_2}$ C) $\frac{R}{-\mu_1 + \mu_2}$ D) $\frac{\mu_1 R}{\mu_2}$

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18. A monochromatic beam of light has a frequency $v=\frac{3}{2\pi}\times10^{12}$ Hz and is propagating along the direction $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$. It is polarized along the $\hat{k}$ direction. The acceptable form for the magnetic field is

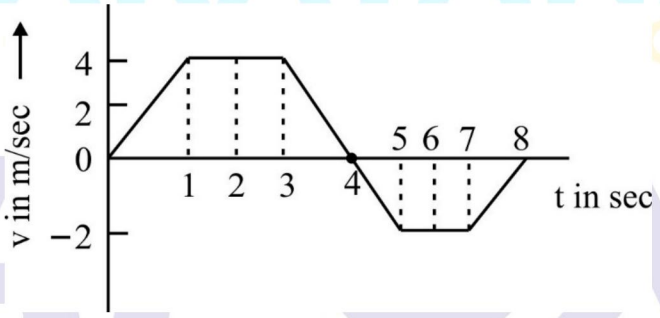
A) $\frac{E_0}{C}\frac{\left(\hat{i}+\hat{j}+\hat{k}\right)}{\sqrt{3}}\cos\left[10^4\frac{\left(\hat{i}+\hat{j}\right)}{\sqrt{2}}.\bar{r}+\left(3\times10^{12}\right)t\right]$

B) $\frac{E_0}{C}\frac{\left(\hat{i}-\hat{j}\right)}{\sqrt{2}}\cos\left[10^4\frac{\left(\hat{i}+\hat{j}\right)}{\sqrt{2}}.\bar{r}-\left(3\times10^{12}\right)t\right]$

C) $\frac{E_0}{C}\frac{\left(\hat{i}-\hat{j}\right)}{\sqrt{2}}\cos\left[10^4\frac{\left(\hat{i}-\hat{j}\right)}{\sqrt{2}}.\bar{r}-\left(3\times10^{12}\right)t\right]$

D) $\frac{E_0}{C}\hat{k}\cos\left[10^4\frac{\left(\hat{i}+\hat{j}\right)}{\sqrt{2}}.\bar{r}+\left(3\times10^{12}\right)t\right]$

19. The velocity versus time graph of a linear motion is shown in the figure. The distance from the origin after 8 sec is



A) 18m B) 16 m C) 8 cm D) 6 m

20. Four equal positive charges, each of value Q, are arranged at the corners of a square of diagonal 2a. A small body of mass m carrying a unit positive charge is placed at a height h above the center of the square (perpendicular to plane). What should be the value of Q in order that this body may be in equilibrium?

A) $\pi\in_0\frac{mg}{2h}\left(h^2+2a^2\right)^{\frac{3}{2}}$ B) $\pi\in_0\frac{mg}{h}\left(h^2+a^2\right)^{\frac{3}{2}}$

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C) $\pi \in_0 \frac{mg}{h}\left(h^2+2a^2\right)^{\frac{3}{2}}$ D) $\pi \in_0 \frac{mg}{2h}\left(h^2-2a^2\right)^{\frac{3}{2}}$

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
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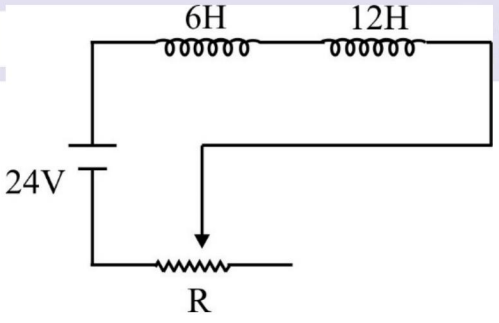
21. Consider a solid sphere of radius R with centre located at the origin. A spherical cavity of a radius $\frac{R}{3}$ is removed from this in such a way that the edge of the cavity touches the edge of the sphere The distance of centre of mass of residual or remaining sphere from the origin will $\frac{R}{N}$; $N=?$

22. If $\bar{P}=m\hat{i}+3\hat{j}-2\hat{k}$ and $\bar{Q}=\hat{i}-2\hat{j}+2\hat{k}$ are mutually perpendicular, then the value of $'m'$ is____

23. A Di-atomic gas initially at 27^0C temperature is compressed suddenly to $\frac{1}{32}$ of its initial volume. The temperature after compression will be ____K (approx)

24. An object of mass m is projected at an angle 60^0 with the horizontal and with velocity u. Its kinetic energy, when it has completed horizontal distance equal to one-third of its range, is $\frac{1}{k}mu^2$, then the value of $'k'$ is____

25. In the given circuit the sliding contact is pulled outwards such that electric current in the circuit changes at the rate of 2 A/s. At an instant when R is 10Ω, the value of current in the circuit will be ____A (neglect mutual induction)



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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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26. Symbol of the Element that does not belong to same group of the remaining elements (Modern periodic table) is

A) Pr B) Pm C) Pa D) Pd

27. If n_1 and n_2 are the boundary value principal quantum numbers of a portion of spectrum of emission spectrum of H atom, the wavelength (in metre) corresponding to last line (longest λ) is.

Given $n_1+n_2=7$, $n_2-n_1=3$ and $R_H=1.097\times10^7 m^{-1}$. (Give your answer in multiple of 10^{-6})

A) 8×10^{-6} B) 4×10^{-6} C) 10×10^{-6} D) 15×10^{-6}

28. Which of the following orders are correct as per the properties mentioned against each?

i) $Al_2O_3 < SiO_2 < P_2O_3 < SO_2$ Acid strength

ii) $AsH_3 < PH_3 < NH_3$ reducing power

iii) $S < O < Cl < F$ More negative electron gain enthalpy

iv) $H_2O > H_2S > H_2Se > H_2Te$ Thermal stability

A) 1,3 B) 1,2,3 C) 1,4 D) 1,2,4

29. A complex is represented as $CoCl_3xNH_3$. Its 0.1 molal solution in water has $\Delta T_f = 0.558K$, K_f for H_2O is $1.86K\ \ molality^{-1}$. Assuming 100% ionization complex and co-ordination number of Co is six, formula of complex is

A) $\left[Co\left(NH_3\right)_6\right]Cl_3$ B) $\left[Co\left(NH_3\right)_5 Cl\right]Cl_2$

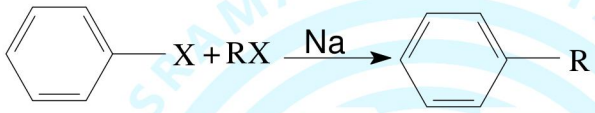
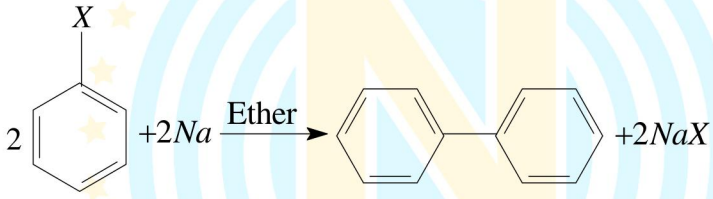
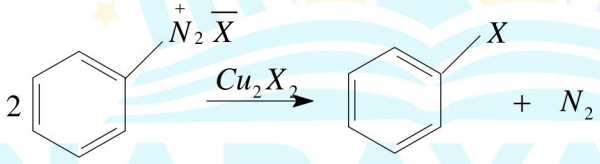
C) $\left[Co\left(NH_3\right)_4 Cl_2\right]Cl$ D) $\left[Co\left(NH_3\right)_3 Cl_3\right]$

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30. The standard reduction potential of Cu^{2+}/Cu and Cu^{2+}/Cu^{+} are 0.337V and 0.153V respectively. The standard electrode potential of Cu^{+}/Cu half-cell is

A) 0.184 V B) 0.827 V C) 0.521 V D) 0.490 V

31. Match the reactions given in column I with the names given in Column II

	Column-I		Column-II
i)		a)	Fittig reaction
ii)		b)	Wurtz Fittig reaction
iii)		c)	Finkelstein reaction
iv)	$C_2H_5Cl + NaI \xrightarrow{\text{dry acetone}} C_2H_5I + NaCl$	d)	Sandmeyer reaction

A) $i \to a, ii \to b, iii \to d, iv \to c$ B) $i \to b, ii \to a, iii \to d, iv \to c$

C) $i \to a, ii \to b, iii \to c, iv \to d$ D) $i \to b, ii \to a, iii \to c, iv \to d$

32. Statement – I: the pentaacetate of glucose does not react with hydroxyl amine indicating the absence of free aldehyde group.

Statement – II: Glucose on oxidation with nitric acid gives glutaric acid

A) Statement-I is False and Statement-II is True

B) Statement-I is True and Statement-II is True

C) Statement-I is True and Statement-II is False

D) Statement-I is False and Statement-II is False

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33. From 392 mg of H_2SO_4, 1.204×10^{21} molecules of H_2SO_4 are removed. Number of moles of H_2SO_4 left are

A) 2×10^{-3} B) 1.2×10^{-3} C) 4×10^{-3} D) 1.5×10^{-3}

34. Calculate the entropy change when 3.6 g of liquid water is completely converted into vapour at 100^0C. The molar heat of vaporization is 40.85 kJ mol^{-1}

A) 6.08 JK^{-1} B) $-21.89J\ K^{-1}$ C) $21.89J\ K^{-1}$ D) $109.5J\ K^{-1}$

35. The magnetic moment (spin only) and hybridization of central atom in $\left[MnBr_4\right]^{-2}$ are

A) $1.73BM, dsp^2$ B) $1.73BM, sp^3$

C) $5.9BM, sp^3$ D) $5.9BM, dsp^2$

36. Which of the following is incorrect statement

A) $\left[Co\left(NH_3\right)_4Cl_2\right]^+$ exhibits geometrical isomerism

B) $Cis-\left[CrCl_2\left(OX\right)_2\right]^{-3}$ is optically inactive

C) $\left[Co\left(NH_3\right)_3\left(NO_2\right)_3\right]$ exhibits facial and meridional isomerism

D) In a co-ordination entity of the type, $\left[Pt\left(Cl_2\right)\left(en\right)_2\right]^{+2}$ only the *Cis* isomer shows optical activity

37. Identify the option which represents the correct products of the following reaction

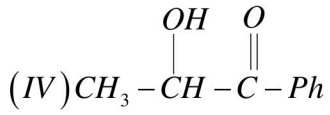
$PhCHO+CH_3CHO\xrightarrow{OH^-}\left(Aldols\right)$

$(I)\ Ph-\underset{\underset{OH}{|}}{CH}-CH_2-CHO$

$(II)\ PhCH_2-OCOPh$

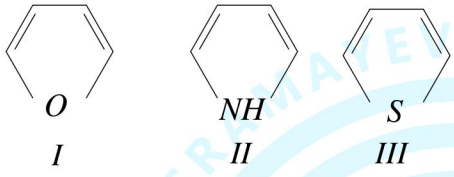
$(III)\ CH_3-\underset{\underset{OH}{|}}{CH}-CH_2CHO$

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A) I, II B) I, III C) II, IIII D) I, III, IV

38. The correct order of resonance energies of the following compounds is___

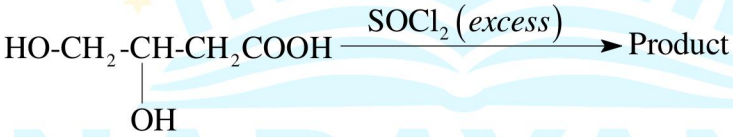


A) $II > III > I$ B) $I > III > II$ C) $II > I > III$ D) $III > II > I$

39. Number of dichloro cyclopentanes (including stereoisomers) formed when cyclopentane reacts with excess of chlorine at high temperature?

A) 7 B) 5 C) 4 D) 9

40.



What will be the molar mass in gram of the organic product of above reaction?

A) 105.5 B) 175.5 C) 138.5 D) 69.5

41. Which of the following set of ions has different colours

A) $Sc^{3+}, Ti^{4+}, Zn^{2+}$ B) Fe^{2+}, Ni^{2+}, V^{3+}

C) V^{4+}, Fe^{3+}, Mn^{2+} D) V^{2+}, Cr^{3+}, Mn^{3+}

42. Statement-I: In Lassaigne's test, the organic compound is fused with a piece of sodium metal

Statement-II: In Lassaigne's test, an ionic compound is converted into covalent compound

A) Statement-I is False and Statement-II is True

B) Statement-I is True and Statement-II is True

C) Statement-I is False and Statement-II is False

D) Statement-I is True and Statement-II is False

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43. A solution which is 0.1 M in Nal and also 0.1 M in Na_2SO_4 is treated with solid $Pb(NO_3)_2$. Compound that is precipitated first and the concentration of anion of the least soluble compound when the more soluble one starts precipitating is (Given $K_{sp}(PbI_2)=9\times10^{-9}$, $K_{sp}(PbSO_4)=1.8\times10^{-8}$

Assume no hydrolysis of Pb^{2+} ion)

A) $PbSO_4, 0.05M$ B) $PbSO_4, 0.02M$

C) $PbI_2, 0.05M$ D) $PbI_2, 0.02M$

44. Consider the following and match the given table

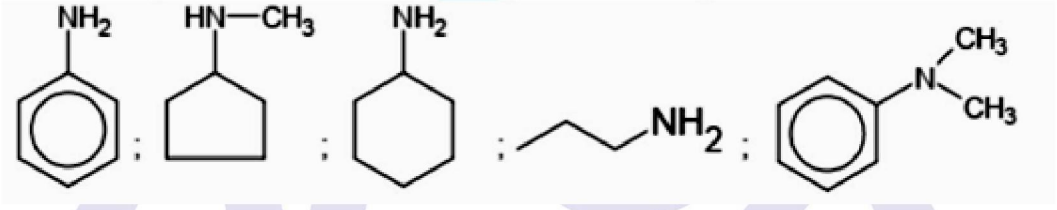
$BCl_3, XeF_2, NO_2, IF_7, CH_4, NO, ClF_3, CO_2, ClO_2, AlCl_3$

1) Molecules with odd electrons	p) CH_4, CO_2
2) Molecules with Super octet	q) NO, NO_2, ClO_2
3) Molecules with octet	r) $BCl_3, AlCl_3$
4) e^- deficient molecules	s) XeF_2, ClF_3, IF_7

A) $1\to r, 2\to s, 3\to p, 4\to q$ B) $1\to p, 2\to s, 3\to q, 4\to r$

C) $1\to p, 2\to r, 3\to q, 4\to s$ D) $1\to q, 2\to s, 3\to p, 4\to r$

45. Number of compounds that can give positive carbyl amine test is



A) 1 B) 4 C) 3 D) 5

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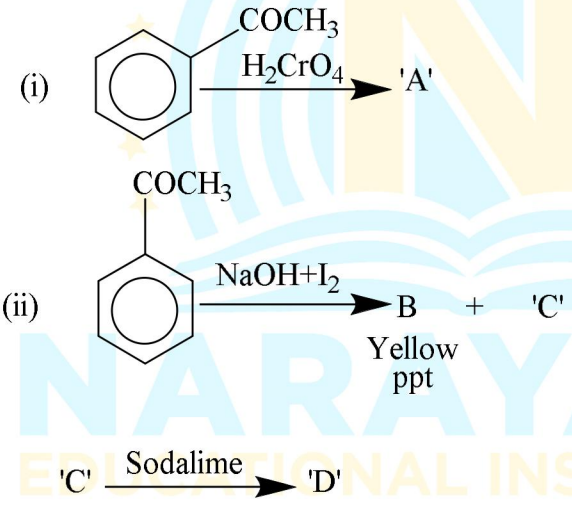
SECTION – II
(Numerical Value Answer Type)

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46. The value of Kb for $0.1M$ solution of mono acidic base having P^H of 10.48 is $x\times10^{-7}$, then the value of $'x'$ is___ (nearest integer, log3=0.48)

47. In sulphur estimation, 0.157 g of an organic compound gave 0.4813 g of barium sulphate. The percentage of sulphur in the compound is_____(nearest integer)

48.



The number of π bonds in the compound $'A' = x$

The number π bonds in the compound $'D' = y$

The value of $x + y$ is_____

49. The gas phase decomposition of dimethyl ether follows first order kinetics

$$CH_3OCH_3(g) \rightarrow CH_4(g) + H_2(g) + CO(g)$$

The reaction is carried out in a constant volume container at 500^0C and has a half life of 14.5 minutes. Initially only dimethyl ether is present at a pressure of 0.40 atmosphere. Total pressure of the system after 12 minutes is_____ (Give your answer by multiplying with 100)

(Antilog 0.25 = 1.778)

For More Join: @IITJEE_Advanc

50. For the reaction at $25^0C, X_2O_4(\ell) \rightarrow 2XO_2(g)$ $\Delta H = 2.1$ K.cal and $\Delta S = 20$ cal K^{-1}. ΔG for the reaction in calories is $-'X'$, the value of $'X'$ is___

MATHEMATICS MAX.MARKS: 100

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(SINGLE CORRECT ANSWER TYPE)

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51. $\int_{\pi/6}^{\pi/3} \frac{3\tan^2 x + 6\tan x + 11}{1+\tan^2 x} dx = = \frac{\pi a + b}{6}$ then $a+b=$

A) 10 B) 12 C) 16 D) 14

52. If $\int \frac{dx}{\sqrt[4]{(x-1)^3(x+2)^5}} = k\sqrt[4]{\frac{x-1}{x+2}} + C$ then $3k=$

A) $\frac{1}{4}$ B) 4 C) 9 D) 1

53. Let f be a function defined on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ as follows

$f(x) = \begin{cases} \frac{2^{|x|}.e^{|x|} - |x| - |x|\ln 2 - 1}{x\tan x} & , x \neq 0 \\ k & , x = 0 \end{cases}$ the value of k so that f is continuous at $x=0$ is

A) $\frac{1}{2}(\ln 2)^2 + \frac{1}{2}\ln 2 + 1$ B) $(\ln 2)^2 + \frac{1}{2}(\ln 2) + 1$

C) $(\ln 2)^2 + (\ln 2) + \frac{1}{2}$ D) $\frac{1}{2}(\ln 2)^2 + \ln 2 + \frac{1}{2}$

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54. Locus of a point P which divides all chords of slope $\frac{1}{2}$ of parabola $x^2=4y$ in the ratio 1:2 internally is another parabola with vertex $\left(\frac{a}{9},\frac{b}{9}\right)$ and length of latus rectum is $\frac{c}{9}$, then the value of $a.b.c$ is___

A) 64 B) 74 C) 100 D) 50

55. If a solution of the differential equation $\frac{dy}{dx}=\frac{1}{x\cos y+\sin 2y}$ is $x=c.e^{\sin y}-k\left(1+\sin y\right)$ then the value of k is __

A) 2 B) 5 C) -2 D) -5

56. Let $f\left(x\right)=x^5+2e^{x/4}$ for all $x\in R$ consider a function $g\left(x\right)$ such that $gof\left(x\right)=x$ for all $x\in R$ then the value of $8.g^1\left(2\right)=$

A) 16 B) 4 C) 8 D) 2

57. Let O be an interior point of ΔABC such that $\overrightarrow{OA}+2\overrightarrow{OB}+3\overrightarrow{OC}=\overrightarrow{O}$. Then the ratio of area of ΔABC to the area of ΔAOC is____

A) 2 B) 1.5 C) 3 D) $\frac{5}{3}$

58. In an $AP, t_1,t_2,t_3.....t_m,$ if $t_1+t_4+t_8+t_{12}+t_{16}+t_{20}+t_{24}+t_{27}=224,$ then $t_1+t_2+t_3+.....+t_{27}$ is equal to____

A) 576 B) 448 C) 675 D) 756

59. Let P be the point $\left(10,-2,-1\right)$ and Q be the foot of the perpendicular drawn from the point $R\left(1,7,6\right)$ on the line passing through the points $\left(2,-5,11\right)$ and $\left(-6,7,-5\right)$. Then the length of line segment PQ is equal to

A) 14 B) 13 C) 12 D) 17

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60. The least value of $|z|$ where z is a complex number which satisfies the inequality

$$\exp\left(\frac{(|z|+3)(|z|-1)}{|z|+1}.\log_e 2\right) \geq \log_{\sqrt{2}}\left|5\sqrt{7}+9i\right|,\ i=\sqrt{-1}\ ,\text{ is equal to}$$

A) 3 B) $\sqrt{5}$ C) 2 D) 8

61. Let $A=\{1,2,3,4,5\}$. Let R be a relation on A defined by xRy, if and only if $4x \leq 5y$. Let m be the number of elements in R and n be the minimum number of elements from $A\times A$ that are required to be added to R to make it a symmetric relation then $m+n$ is equal to

A) 24 B) 25 C) 26 D) 23

62. Number of ways in which letters of the word 'HONGKONG' can be arranged such that O and N never come together, is____

A) 1440 B) 1320 C) 960 D) 600

63. Let $P(5,4), Q(-2,4)$ and $R(a,b)$ be the vertices of a triangle, and $H\left(2,\frac{14}{5}\right)$ be the orthocentre of ΔPQR then $2a-b=$

A) 19 B) 11 C) 10 D) 12

64. The value of x satisfying the equation

$$\left(\sin^{-1}x\right)^3-\left(\cos^{-1}x\right)^3+\left(\sin^{-1}x\right)\left(\cos^{-1}x\right)\left(\sin^{-1}x-\cos^{-1}x\right)=\frac{\pi^3}{16}$$ can be equal to

A) $\cos\frac{\pi}{5}$ B) $\cos\frac{\pi}{8}$ C) $\cos\frac{\pi}{4}$ D) $\cos\frac{\pi}{12}$

65. If $P=\sqrt{q^2\cos^2\theta+r^2\sin^2\theta}+\sqrt{q^2\sin^2\theta+r^2\cos^2\theta}$ then the difference between maximum and minimum values of P^2 is given by.... $\left(\forall q,r\in R^+\right)$

A) $(q-r)^2$ B) $(q+r)^2$ C) q^2+r^2 D) q^2-r^2

66. Let sets A and B have 5 elements each let the mean of the elements in sets A and B be 5 and 8 respectively and the variance of the elements in sets A and B be 12 and 20 respectively. A new set of 10 elements is formed by subtracting 3 from each element of A and adding 2 to each element of B. Then the sum of the mean and variance of the elements of C is

A) 38 B) 40 C) 36 D) 32

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67. Find the volume of the tetrahedron whose vertices are $A(1,1,0), B(-4,3,6,)$, $C(-1,0,3)$ and $D(2,4,-5)$

A) 6 B) 36 C) 12 D) 18

68. Let A be a matrix of order 3×3 and $|A|=2$, the value of $\left|3adj\left(2A^{-1}.adj\left(2A^{-1}\right)\right)\right|$ is $3^{\alpha}.2^{\beta}$ then $\alpha+\beta=$

A) 22 B) 21 C) 15 D) 36

69. The system of homogeneous equations $\lambda x+(\lambda+1)y+(\lambda-1)z=0$, $(\lambda+1)x+\lambda y+(\lambda+2)z=0, (\lambda-1)x+(\lambda+2)y+\lambda z=0$ has non trivial solution for

A) Exactly 3 real values of λ

B) Exactly 2 real values of λ

C) Exactly 1 real value of λ

D) Infinitely many real values of λ

70. A,B,C in order roll a fair die indefinitely until a winner is obtained. A wins if he rolls a prime number, B wins if he rolls a composite number whereas C wins if he rolls neither prime nor composite. Let $P(A),P(B),P(C)$ denotes their respective chances of winning then identify the correct statement

A) $P(A)+P(B)=\frac{12}{13}$ B) $P(B)+P(C)=\frac{2}{3}$

C) $P(A)=5P(C)$ D) $P(A)+P(C)=\frac{5}{13}$

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. Let $f(x)=\max\{|x+1|,|x+2|,.....|x+5|\}$. Then $\int_{-6}^{0} f(x)\,dx$ is

72. If the constant term in the expansion of $\left(x^3+5x^2-\frac{2}{x^5}\right)^{10}$ is $-\left(2^{\alpha_1}.3^{\alpha_2}.5^{\alpha_3}.7^{\alpha_4}\right)$ then $\alpha_1\alpha_2+\alpha_3\alpha_4$ is equal to

73. Let C be a curve which is locus of point of intersection of lines $x=2+m$ and $my=4-m$. A circle $S\equiv(x-2)^2+(y+1)^2=25$ intersects the curve C at four point P,Q,R and S. If 'O' is the centre of the curve 'C' then $OP^2+OQ^2+OC^2+OR^2$ is equal to _____

74. The function $f(x)=\frac{x}{x^2-6x-16}, x\in R-\{-2,8\}$ decreases in $(-\infty,-a)\cup(-a,b)\cup(b,\infty)$ then $a+b$ is equal to

75. The value of $6+\log_{3/2}\left(\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}.....}}}\right)$ _____ is equal to

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